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День 3

Раздел 1

1.

```
root@eltex-practice2-v22:~# su - user1
user1@eltex-practice2-v22:~$
```

2.

```
user1@eltex-practice2-v22:~$ ps -eo nlwp --no-headers | uniq -c
  69    1
   1    7
   6    1
   1    2
   3    1
   1    4
   1    2
   1    1
   1    6
   1    1
   1    4
   1    2
   1    4
   1    6
   1    4
  22    1
user1@eltex-practice2-v22:~$
```

3.

```
top - 11:52:01 up 35 min,  1 user,  load average: 0,68, 0,69, 0,60
Threads: 1186 total, 1 running, 1185 sleep, 0 d-sleep, 0 stopped, 0 zombie
%Cpu(s):  3,1 us,  3,5 sy,  0,0 ni, 92,4 id,  0,0 wa,  0,7 hi,  0,2 si,  0,0 st
MiB Mem : 7263,1 total,  455,7 free, 4130,2 used, 2776,1 buff/cache
MiB Swap: 8192,0 total,  8192,0 free,  0,0 used. 3133,0 avail Mem
```

PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND
376	1	975	systemd+	systemd+	?	0:00.06	0,0	0,1	S	systemd-timesyn
382	1	975	systemd+	systemd+	?	0:00.00	0,0	0,1	S	sd-resolve
1051	1	133	rtkit	rtkit	?	0:00.03	0,0	0,0	S	rtkit-daemon
1060	1	133	rtkit	rtkit	?	0:00.00	0,0	0,0	S	rtkit-daemon

4.

```
eltex-v22@eltex-2026-summer-12:10>passwd
Changing password for eltex-v22.
Current password: 
```

5.

```
Cannot find terminfo entry for 'xterm-kitty'.
```

```
root@eltex-practice2-v22:~# top
```

```
root@eltex-practice2-v22:~# screet
```

```
Command 'screet' not found, did you mean:
```

```
  command 'screed' from deb python3-screed (1.1.2-1)
```

```
  command 'screen' from deb screen (4.9.1-1ubuntu1)
```

```
Try: apt install <deb name>
```

```
root@eltex-practice2-v22:~# exiy
```

```
Command 'exiy' not found, did you mean:
```

```
  command 'exim' from deb exim4-daemon-heavy (4.97-4ubuntu4.6)
```

```
  command 'exim' from deb exim4-daemon-light (4.97-4ubuntu4.6)
```

```
  command 'exip' from deb exip (0.1.0-2ubuntu0.24.04.2)
```

```
  command 'exif' from deb exif (0.6.22-3)
```

```
Try: apt install <deb name>
```

```
root@eltex-practice2-v22:~# exit
```

```
logout
```

```
Connection to 172.16.9.196 closed.
```

```
eltex-v22@eltex-2026-summer-12:06>screen -l
```

```
bash: screen: command not found
```

```
eltex-v22@eltex-2026-summer-12:06>screen -l
```

```
[screen is terminating]
```

```
eltex-v22@eltex-2026-summer-12:06>passwd
```

```
Changing password for eltex-v22.
```

```
Current password: 
```

```
eltex-2026-summer: screen -l
```

```
top - 12:21:54 up 14:21, 38 users,  load average: 0.10, 0.07, 0.01
Tasks: 319 total,  2 running, 311 sleeping,   6 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  3914.7 total,  2706.0 free,   753.7 used,   740.7 buff/cache
MiB Swap:   976.0 total,   976.0 free,    0.0 used.  3161.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
11348	root	20	0	8780	3504	3180	S	0.0	0.1	0:00.00	passwd ←
11317	root	20	0	17844	11180	9584	S	0.0	0.3	0:00.00	sshd
11316	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/3:1
11311	root	20	0	0	0	0	I	0.0	0.0	0:00.01	kworker/1:0-ata_sff
11256	root	20	0	0	0	0	I	0.0	0.0	0:00.06	kworker/u8:2-events_unbound
11255	root	20	0	0	0	0	I	0.0	0.0	0:00.09	kworker/u8:1-events_unbound
11195	root	20	0	0	0	0	I	0.0	0.0	0:00.01	kworker/1:1-ata_sff
11145	root	20	0	17756	10872	9264	S	0.0	0.3	0:00.03	sshd

```
MiB Mem :  3914.7 total,  2745.7 free,   714.3 used,   739.8 buff/cache
```

```
MiB Swap:   976.0 total,   976.0 free,    0.0 used.  3200.4 avail Mem
```

```
Send pid 11348 signal [15/sigterm]
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
11348	root	20	0	8780	3504	3180	S	0.0	0.1	0:00.00	passwd

```
eltex-v22@eltex-2026-summer-12:06>passwd
Changing password for eltex-v22.
Current password: Killed
eltex-v22@eltex-2026-summer-12:06>|
```

```
eltex-2026-summer: screen -l

kill: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l
Send a signal to a job.

Send the processes identified by PID or JOBSPEC the signal named by
SIGSPEC or SIGNUM. If neither SIGSPEC nor SIGNUM is present, then
SIGTERM is assumed.

Options:
  -s sig      SIG is a signal name
  -n sig      SIG is a signal number
  -l          list the signal names; if arguments follow '-l' they are
              assumed to be signal numbers for which names should be
              listed
  -L          synonym for -l

Kill is a shell builtin for two reasons: it allows job IDs to be used
instead of process IDs, and allows processes to be killed if the limit
on processes that you can create is reached.

Exit Status:
Returns success unless an invalid option is given or an error occurs.

eltex-v22@eltex-2026-summer-12:11>kill -s 2 11348
eltex-v22@eltex-2026-summer-12:11>kill -s 3 11348
eltex-v22@eltex-2026-summer-12:11>kill -s 9 11348
eltex-v22@eltex-2026-summer-12:11>|
```

6.

```
eltex-v22@eltex-2026-summer-12:06>vim ~/file_task3.txt

[1]+  Stopped                  vim ~/file_task3.txt
eltex-v22@eltex-2026-summer-12:06>
```

7.

```
eltex-v22@eltex-2026-summer-12:06>sleep 600
^Z
[2]+  Stopped                  sleep 600
eltex-v22@eltex-2026-summer-12:06>jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
eltex-v22@eltex-2026-summer-12:06>
```

8.

```
eltex-v22@eltex-2026-summer-12:06>sleep 600 &
[3] 11950
eltex-v22@eltex-2026-summer-12:06>jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
[3]   Running                  sleep 600 &
eltex-v22@eltex-2026-summer-12:06>
```

9.

```
eltex-v22@eltex-2026-summer-12:06>renice -n 10 -p 11950
11950 (process ID) old priority 0, new priority 10
eltex-v22@eltex-2026-summer-12:06>
```

10.

```
eltex-v22@eltex-2026-summer-12:06>ps -o pid,comm,nice -p 11950
  PID COMMAND      NI
  11950 sleep        10
[2]   Done          sleep 600
eltex-v22@eltex-2026-summer-12:06>
```

11.

```
eltex-v22@eltex-2026-summer-12:06>jobs
[1]+  Stopped                  vim file_task3.txt
eltex-v22@eltex-2026-summer-12:06>jobs -l
[1]+ 12565 Stopped              vim file_task3.txt
eltex-v22@eltex-2026-summer-12:06>jobs 12565
bash: jobs: 12565: no such job
eltex-v22@eltex-2026-summer-12:06>fg 1
vim file_task3.txt
eltex-v22@eltex-2026-summer-12:06>
```

12.

```
eltex-v22@eltex-2026-summer-12:06>jobs -l
[1]+ 12626 Running              sleep 600 &
eltex-v22@eltex-2026-summer-12:06>kill -s 15 12626
[1]+  Terminated             sleep 600
eltex-v22@eltex-2026-summer-12:06>jobs
eltex-v22@eltex-2026-summer-12:06>
```

13.

```
user1@eltex-practice2-v22:~$ trap 'echo "Меня голыми руками не возьмёшь!"' SIGINT SIGQUIT

user1@eltex-practice2-v22:~$ kill -SIGINT $$
Меня голыми руками не возьмёшь!
user1@eltex-practice2-v22:~$ ^CМеня голыми руками не возьмёшь!

user1@eltex-practice2-v22:~$ kill -SIGQUIT $$
Меня голыми руками не возьмёшь!
user1@eltex-practice2-v22:~$ █
```

Раздел 2.

1,2.

```
user1@eltex-practice2-v22:~$ chmod +x cuckoo.sh
user1@eltex-practice2-v22:~$ ls
cuckoo.sh
user1@eltex-practice2-v22:~$ nohup ./cuckoo.sh &
```

```
user1@eltex-practice2-v22:~$ nohup ./cuckoo.sh &
[1] 5565
user1@eltex-practice2-v22:~$ nohup: ignoring input and appending output to 'nohup.out'

user1@eltex-practice2-v22:~$ jobs
[1]+  Running                  nohup ./cuckoo.sh &
user1@eltex-practice2-v22:~$ cat cockoo.log
2026-07-11 05:47:06 Startup!
user1@eltex-practice2-v22:~$ echo "DaNil[1245]: how much time do I have?" > /tmp/run/cuckoo.pid
user1@eltex-practice2-v22:~$ cat cockoo.log
2026-07-11 05:47:06 Startup!
2026-07-11 05:47:30 DaNil[1245] 6
user1@eltex-practice2-v22:~$ jobs -l
[1]+  5565 Running                  nohup ./cuckoo.sh &
user1@eltex-practice2-v22:~$ kill -TERM 5565
user1@eltex-practice2-v22:~$ jobs -l
[1]+  5565 Done                    nohup ./cuckoo.sh
user1@eltex-practice2-v22:~$ cat cockoo.log
2026-07-11 05:47:06 Startup!
2026-07-11 05:47:30 DaNil[1245] 6
2026-07-11 05:47:53 Shutdown!
user1@eltex-practice2-v22:~$
```

```
user1@eltex-practice2-v22:~$ cat cuckoo.sh
#!/bin/bash

FIFO_DIR="/tmp/run/"
FIFO_NAME="cuckoo.pid"
FIFO_PATH="$FIFO_DIR/$FIFO_NAME"
LOG_FILE="cockoo.log"

log() {
    echo "$(date '+%Y-%m-%d %H:%M:%S') $1" >> "$LOG_FILE"
}

log_and_answer() {
    echo "$(date '+%Y-%m-%d %H:%M:%S') $1" >> "$LOG_FILE"
    echo "$(date '+%Y-%m-%d %H:%M:%S') $1" >> "$FIFO_PATH"
}

mkdir -p "$FIFO_DIR"

rm -f "$FIFO_PATH"
rm -f "$LOG_FILE"

mkfifo "$FIFO_PATH"

log "Startup!"

ending() {
    log "Shutdown!"
    rm -f "$FIFO_PATH"
    exit 0
}

# При получении сигнала завершает работу
trap ending SIGTERM

while true; do
    if read -r line < "$FIFO_PATH"; then
        if [[ "$line" =~ .*\[([0-9]*):\ how\ much\ time\ do\ I\ have\?$ ]]; then
            NAME=$(echo "$line" | awk -F [ '{print $1}')
            PID=$(echo "$line" | awk -F [ '{print $2}' | awk -F ] '{print $1}')
            N=$((RANDOM % 9 + 2))
            log_and_answer "$NAME[$PID] $N"
        fi
    fi
done
user1@eltex-practice2-v22:~$
```

3.

```
user1@eltex-practice2-v22:~$ ls
cockoo.log  cuckoo.sh  nohup.out  other.sh  template_task.sh
user1@eltex-practice2-v22:~$ jobs
[1]+  Running                  nohup ./cuckoo.sh &
user1@eltex-practice2-v22:~$ ./other.sh
user1@eltex-practice2-v22:~$ ls
cockoo.log  cuckoo.sh  nohup.out  other.sh  report_other.log  template_task.sh
user1@eltex-practice2-v22:~$ cat report_other.log
2026-07-11 07:53:36 [6834] Скрипт запущен.
2026-07-11 07:53:42 [6834] Скрипт завершился, работал 6 секунд.
user1@eltex-practice2-v22:~$
```

```
user1@eltex-practice2-v22:~$ cat template_task.sh
#!/bin/bash

NAME=$(echo "$0" | sed 's/.*\///g')

if [[ $NAME =~ template_task.sh ]]; then
    echo "я бригадир, сам не работаю"
    exit 0
fi

SHORT_NAME=$(echo $NAME | awk -F . '{print $1}')

LOG_NAME="report_${SHORT_NAME}.log"

echo "$(date '+%Y-%m-%d %H:%M:%S') [$$] Скрипт запущен." >> "$LOG_NAME"

echo "$SHORT_NAME[$$]: how much time do I have?" > /tmp/run/cuckoo.pid

read -r MESSAGE < /tmp/run/cuckoo.pid

N=$(echo "$MESSAGE" | awk '{print $4}')

sleep "$N"

echo "$(date '+%Y-%m-%d %H:%M:%S') [$$] Скрипт завершился, работал $N секунд." >> "$LOG_NAME"
user1@eltex-practice2-v22:~$ ./template_task.sh
я бригадир, сам не работаю
user1@eltex-practice2-v22:~$
```

4.

```
user1@eltex-practice2-v22:~$ cat observer.sh
#!/bin/bash

CONF_FILE="/home/user1/observer.conf"

LOG_FILE="/home/user1/observer.log"

find_proc() {
    for pid in /proc/[0-9]*; do
        if [ -r "$pid/comm" ]; then
            if grep -qx "$1" "$pid/comm" 2>/dev/null; then
                return 0
            fi
        fi
    done
    return 1
}

while read line; do
    FILE_PATH="/home/user1/$line"
    if ! find_proc "$FILE_PATH"; then
        nohup "$FILE_PATH" &
        echo "restarting... $line" >> "$LOG_FILE"
    fi
done < "$CONF_FILE"
user1@eltex-practice2-v22:~$
```

5.

```
# m h dom mon dow  command
* * * * * /home/user1/observer.sh
```


6.

```
user1@eltex-practice2-v22:~$ ls
cockoo.log  observer.conf  other1.sh  report_other1.log  template_task.sh
cuckoo.sh  observer.log   other2.sh  report_other2.log
nohup.out  observer.sh    other.sh   report_other.log
user1@eltex-practice2-v22:~$ cat cockoo.log
2026-07-11 23:40:55 Startup!
2026-07-11 23:42:03 other[10899] 8
2026-07-11 23:42:03 other1[11033] 2
2026-07-11 23:42:03 other2[11169] 6
2026-07-12 00:04:01 other[11603] 4
2026-07-12 00:04:01 other1[11740] 9
2026-07-12 00:04:01 other2[11879] 9
2026-07-12 00:05:01 other[12147] 10
2026-07-12 00:05:01 other1[12284] 8
2026-07-12 00:05:01 other2[12423] 2
2026-07-12 00:06:01 other[12683] 4
2026-07-12 00:06:01 other1[12818] 4
2026-07-12 00:06:01 other2[12955] 9
2026-07-12 00:07:01 other[13236] 2
2026-07-12 00:07:02 other1[13373] 2
2026-07-12 00:07:02 other2[13512] 9
2026-07-12 00:08:01 other[13773] 5
2026-07-12 00:08:01 other1[13910] 2
2026-07-12 00:08:01 other2[14049] 5
2026-07-12 00:09:01 other[14313] 2
2026-07-12 00:09:01 other1[14451] 6
2026-07-12 00:09:01 other2[14591] 9
2026-07-12 00:10:01 other[14858] 8
2026-07-12 00:10:01 other1[14997] 7
2026-07-12 00:10:01 other2[15138] 5
user1@eltex-practice2-v22:~$ cat observer.conf
template_task.sh
other.sh
other1.sh
other2.sh
user1@eltex-practice2-v22:~$
```

```
user1@eltex-practice2-v22:~$ tar -czf scripts.tar.gz *.log *.sh *.conf
user1@eltex-practice2-v22:~$ ls
cockoo.log  observer.conf  other1.sh  report_other1.log  scripts.tar.gz
cuckoo.sh  observer.log   other2.sh  report_other2.log  template_task.sh
nohup.out  observer.sh    other.sh   report_other.log
user1@eltex-practice2-v22:~$
```